THE MULTIPLICITY OF TWO-PERIODIC COMPLEXES IS ADDITIVE IN SHORT EXACT SEQUENCES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The cross-chapter additive-length lemma is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Two-periodic complexes and their multiplicity e_R. Same chapter, Definition environment labelled definition-periodic-complex (source line 110) together with the displayed definition of e_R at source lines 200-205 (label equation-multiplicity-coker-ker, line 203). Supplies the objects (M_i, N_i, phi_i, psi_i), the quantity e_R(M, N, phi, psi) and the meaning of 'cohomology modules of finite length'; the additivity argument is the counted content.
- Additivity of the length function on short exact sequences. Algebra, Lemma labelled algebra-lemma-length-additive (canonical Stacks tag 00IV); invoked at proof lines 252-254. Supplies the numerical additivity used at the very end; the long exact sequence and its truncation into a finite exact sequence with a filtration are inside the counted span.

Source reference. Original lemma 00IV: sources/stacks-chow/tags.txt, source lines 693–693. Cross-chapter standard prerequisite invoked once at proof lines 252-254 for additivity of the length function; it is a numerical input only, and the long exact sequence and its truncation that constitute the counted argument are retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Suppose that we have a short exact sequence of 2-periodic complexes*

$$0 \to (M_1, N_1, \varphi_1, \psi_1) \to (M_2, N_2, \varphi_2, \psi_2) \to (M_3, N_3, \varphi_3, \psi_3) \to 0$$

If two out of three have cohomology modules of finite length so does the third and we have

$$e_R(M_2, N_2, \varphi_2, \psi_2) = e_R(M_1, N_1, \varphi_1, \psi_1) + e_R(M_3, N_3, \varphi_3, \psi_3).$$

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Proof. We abbreviate $A = (M_1, N_1, \varphi_1, \psi_1)$, $B = (M_2, N_2, \varphi_2, \psi_2)$ and $C = (M_3, N_3, \varphi_3, \psi_3)$. We have a long exact cohomology sequence

$$\ldots \to H^1(C) \to H^0(A) \to H^0(B) \to H^0(C) \to H^1(A) \to H^1(B) \to H^1(C) \to \ldots$$

This gives a finite exact sequence

$$0 \to I \to H^0(A) \to H^0(B) \to H^0(C) \to H^1(A) \to H^1(B) \to K \to 0$$

with $0 \to K \to H^1(C) \to I \to 0$ a filtration. By additivity of the length function (Algebra, Lemma 00IV) we see the result. $\square$